



Aitiome

An honest mechanistic-reasoning engine for the
environmental exposome of neurodegeneration.

It recovers the real ones, resolves to the real neurons, and isn't fooled by the imposters.

`aitiome.fly.dev` / Built with Claude: Life Sciences

Non-genetic drivers of sporadic neurodegeneration

Most Parkinson's and Alzheimer's is **sporadic**, not inherited. The environmental exposome - pesticides, metals, industrial chemicals - is a leading suspect, and the field (Miller, Barouki, Samieri; *Nature Neuroscience* 2024) has called for AI that connects chemicals to mechanism.

The hard part isn't listing bioactive chemicals. It's telling a real neurotoxicant apart from a compound that is merely bioactive - the exact failure mode of activity-based screening.

We set out to discover. The evidence redirected us.

FINDING 1

No shippable unsupervised-discovery signal exists for this chemical class on public data. Seven axes tested; all coverage-killed or confounder-killed.

FINDING 2

The discriminating signal is **curated mechanism**, not assay activity. Bioactivity scores worse than chance against the decoys.

So we built the thing the data supports - and made the honesty the contribution.

Three co-equal goals

01

Validation that works

Recover the known neurotoxicants on the endorsed pathway; reject the adversarial decoys.

02

A signature visualization

The recovery-and-specificity reveal resolving into the vulnerable dopaminergic neurons.

03

Honest, calibrated framing

Confidence tiers on every result; the discovery limits shown, not hidden.

THE RESULT, ON THE RECONNAISSANCE GROUND TRUTH

Zero errors

12/12

known neurotoxicants recovered

15/15

negatives correctly rejected

6/6

bioactive decoys not fooled

0

false positives + false negatives

12 positives (6 assay-recovered + 6 curated-anchored) - 15 negatives, including 6 mitochondria-active adversarial decoys.

The recovery rule

```
positive ⇔ ( CTD curated Parkinson's DirectEvidence ) OR ( registered neuro-  
AOP stressor )
```

Curated signals are diagnostic.

Two independent curation efforts. Fixed logical rule, no fitted parameters.

Assay activity is corroboration only.

Anti-diagnostic here - the engine **never** gates a positive call on bioactivity.

VALIDATION, NOT A WATCHLIST

It reconstructs the endorsed mechanism, recovers the known neurotoxicants, and is not fooled by the imposters.

Resolve a chemical: name, CAS, DTXSID, or InChIKey

Resolve

THE SIGNATURE VISUAL

known neurotoxicants

rotenone

paraquat

MPTP

chlorpyrifos

6-hydroxydopamine

The endorsed AOP-3 cascade, grounded, resolving into the vulnerable neurons

MIE complex-I inhibition → mitochondrial dysfunction → nigrostriatal dopaminergic degeneration → parkinsonian deficits. MIE grounded in MitoCarta Complex-I subunits; the terminal frame is the Kamath SOX6/AGTR1^{pa} vulnerable dopaminergic neurons.

It is not fooled by the imposters

WHAT FOOLS AN ACTIVITY MODEL

The decoys are genuinely bioactive, often mitochondria-active. On multi-assay fingerprint similarity they collapse into the positives (0.53).

WHY AITIOME DOES NOT

No curated diagnostic signal. Independently, warfarin and fenofibrate also fail brain-exposure and show zero FAERS parkinsonism signal - rejected on three independent lines at once.

Why not just use bioactivity?

If activity could do the job, some signal would separate the 12 positives from the 6 decoys. None does - every one is at or below chance (0.5).



Curated rule on the same set: **perfect (0 errors)**. The decoys are, if anything, more bioactive than the real neurotoxicants.

Is it circular? Two independent curations converge

$$\begin{array}{ccccc} 8/12 & + & 8/12 & \rightarrow & 12/12 \\ \text{CTD DirectEvidence alone} & & \text{AOP-Wiki stressor alone} & & \text{the rule (CTD or AOP)} \end{array}$$

CTD (toxicogenomics literature) and AOP-Wiki (OECD regulatory) are independent efforts; neither alone suffices, so this is not one source read twice. Both are **0/15** false-positives. And the decoys were selected for bioactivity, not curation status - so rejecting them is not baked in.

Where AI-driven discovery works, and where it does not

LINCS L1000 COVERAGE-KILLED	EPA HTTr COVERAGE-KILLED	EPA HTTP COVERAGE-KILLED	Structure / QSAR CONFOUNDER-KILLED
ToxCast fingerprint CONFOUNDER-KILLED	ComptoxAI / AlzKB COVERAGE-KILLED	Neural-specific subset QUALIFIED LEAD ●	Boltz-2 Q-site CONDITIONAL LEAD ●

Seven axes coverage- or confounder-killed. Two honest, unproven leads remain (reported with explicit N).
Discovery is a map, never a predictor.

Claude explains. It never decides.

Evidence-reasoner (Opus 4.8)

Writes the calibrated, [E#]-cited synthesis. Hard-bounded: the call and tier are fixed inputs it may not change; it cites real strands, invents nothing, never claims causation.

Claude Science

The human-in-the-loop substrate for evidence curation (e.g. epidemiology extraction), snapshotted into the engine.

Dual interface

One engine over HTTP and MCP - a scientist and an external agent query the same tools.

Deterministic core, auditable end to end

- One Go binary embeds the curated data; the recovery decision is **deterministic** - no LLM in the decision path.
- Transport-agnostic service, thin HTTP + MCP adapters (the dual interface).
- DTXSID-first, salt-form-correct identity (the paraquat-dichloride trap, tested).
- Every claim links to the original source (CTD, AOP-Wiki, MitoCarta, Kamath, ToxCast/ICE, FAERS, B3DB).
- `make validate` prints the falsification report; `fp=fn=0` is enforced in tests.
- ~13 MB container, live on fly.io. Reproducible: same input, same output.

Limitations and next steps

- A curated benchmark with an adversarial control (27), not a population generalization study.
- Recovery shares curated provenance with the labels - defended by the independent-source ablation and the orthogonal negatives; it is a sanity check, not the headline.
- Anchored on the Parkinson's / mitochondrial route; Alzheimer's is a scaffold extension.
- **Next:** a larger, blinded neural-specific reference set to power the one qualified lead (AUROC 0.72, currently perm-p 0.155).
- **Next:** run the bounded Boltz-2 Q-site benchmark - physics, not annotation or activity.
- **Next:** extend the endorsed scaffold to an AD hallmark.



Less flashy than the original pitch. Far more trustworthy.

Validated recovery, adversarial specificity proven by falsification, and an honest map of the limits - for a mechanism-first audience, that is the point.

`aitiome.fly.dev` / `github.com/jonradoff/aitiome`